SMASH NILPOTENCE UP TO TORSION OF DIFFERENCES OF RATIONAL POINTS

THE STACKS PROJECT AUTHORS

Editorial note. The displayed expansion of (sum c_1(L_i))^n has coefficient n!, not the printed n^n. Thus the displayed last coefficient should be ((n-1)!)^n n!, still a nonzero integer. The claimed torsion conclusion uses only a nonzero multiple and is unchanged. The original displays are retained.

Editorial note. In the family paragraph, the symbols M_p and M_q refer to the constructed L-prime fibres. This is a visible notation slip, not an additional line bundle.

Editorial note. The norm identity and projective-bundle description have author-marked omitted routine details. They are covered by the explicitly named standard norm and Abel–Jacobi inputs; no alternate source proof is inserted.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Bertini through finitely many smooth points, Chow pullback/pushforward, norms of line bundles, symmetric powers and Abel–Jacobi projective bundles for n>=2g-1, projective-bundle Picard formula, and Euler characteristic in connected proper flat families.

Source reference. Original lemma 0FH6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19346–19346. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0FD5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19205–19205. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0FD6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19206–19206. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0FD9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19209–19209. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BA0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14193–14193. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

Date: stacks-project:weil.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 056R: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6719–6719. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 056T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6721–6721. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 03RO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4937–4937. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 03RP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4938–4938. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02RH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3624–3624. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0B94: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14162–14162. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0B9C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14170–14170. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0B9G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14174–14174. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BD2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14300–14300. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BD3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14301–14301. Stable Stacks tag for an expressly

allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BCY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14296–14296. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0FDX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19232–19232. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02TX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3712–3712. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BXJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14982–14982. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0B9T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14186–14186. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01XT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2532–2532. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1 (Voevodsky)**.** [1] *Let k be a field. Let X be a geometrically irreducible smooth projective scheme over k. Let $x, x' \in X$ be k-rational points. For n large enough the class of the zero cycle*

$$([x] - [x']) \times \ldots \times ([x] - [x']) \in \mathrm{CH}_0(X^n)$$

is torsion.

Lemma 0.2. *Let k be a field. Let X be a geometrically irreducible smooth projective scheme over k. Let $x, x' \in X$ be k-rational points. Let n be an integer invertible in k. Then there exists a finite separable extension k'/k such that the pullback of $[x] - [x']$ to $X_{k'}$ is divisible by n in $\mathrm{CH}_0(X_{k'})$.*

Proof. Let k' be a separable algebraic closure of k. Suppose that we can show the pullback of $[x] - [x']$ to $X_{k'}$ is divisible by n in $\mathrm{CH}_0(X_{k'})$. Then we conclude by Lemma 0FH6. Thus we may and do assume k is separably algebraically closed.

Suppose $\dim(X) > 1$. Let $\mathcal{L}$ be an ample invertible sheaf on X. Set

$$V = \{s \in H^0(X, \mathcal{L}) \mid s(x) = 0 \text{ and } s(x') = 0\}$$

After replacing $\mathcal{L}$ by a power we see that for a general $v \in V$ the corresponding divisor $H_v \subset X$ is smooth away from x and x', see Varieties, Lemmas 0FD5 and 0FD6. To find v we use that k is infinite (being separably algebraically closed). If we choose s general, then the image of s in $\mathfrak{m}_x\mathcal{L}_x/\mathfrak{m}_x^2\mathcal{L}_x$ will be nonzero, which implies that H_v is smooth at x (details omitted). Similarly for x'. Thus H_v is smooth. By Varieties, Lemma 0FD9 (applied to the base change of everything to the algebraic closure of k) we see that H_v is geometrically connected. It suffices to prove the result for $[x]-[x']$ seen as an element of $\mathrm{CH}_0(H_v)$. In this way we reduce to the case of a curve.

Assume X is a curve. Then we see that $\mathcal{O}_X(x-x')$ defines a k-rational point g of $J = \underline{\mathrm{Pic}}^0_{X/k}$, see Picard Schemes of Curves, Lemma 0BA0. Recall that J is a proper smooth variety over k which is also a group scheme over k (same reference). Hence J is geometrically integral (see Varieties, Lemma 056R and 056T). In other words, J is an abelian variety, see Groupoids, Definition 03RO. Thus $[n] : J \to J$ is finite étale by Groupoids, Proposition 03RP (this is where we use n is invertible in k). Since k is separably closed we conclude that $g = [n](g')$ for some $g' \in J(k)$. If $\mathcal{L}$ is the degree 0 invertible module on X corresponding to g', then we conclude that $\mathcal{O}_X(x-x') \cong \mathcal{L}^{\otimes n}$ as desired. □

Lemma 0.3. *Let K/k be an algebraic extension of fields. Let X be a finite type scheme over k. The kernel of the map $\mathrm{CH}_i(X) \to \mathrm{CH}_i(X_K)$ constructed in Lemma 0FH6 is torsion.*

Proof. It clearly suffices to show that the kernel of flat pullback $\mathrm{CH}_i(X) \to \mathrm{CH}_i(X_{k'})$ by $\pi : X_{k'} \to X$ is torsion for any finite extension k'/k. This is clear because $\pi_*\pi^*\alpha = [k' : k]\alpha$ by Chow Homology, Lemma 02RH. □

Proof. If we can show this after base change to the algebraic closure of k, then the result follows over k because the kernel of pullback is torsion by Lemma 0.3. Hence we may and do assume k is algebraically closed.

Using Bertini we can choose a smooth curve $C \subset X$ passing through x and x'. See proof of Lemma 0.2. Hence we may assume X is a curve.

Assume X is a curve and k is algebraically closed. Write $S^n(X) = \underline{\mathrm{Hilb}}^n_{X/k}$ with notation as in Picard Schemes of Curves, Sections 0B94 and 0B9C. There is a canonical morphism

$$\pi : X^n \longrightarrow S^n(X)$$

which sends the k-rational point $(x_1, \ldots, x_n)$ to the k-rational point corresponding to the divisor $[x_1]+\ldots+[x_n]$ on X. There is a faithful action of the symmetric group S_n on X^n. The morphism π is S_n-invariant and the fibres of π are S_n-orbits (set theoretically). Finally, π is finite flat of degree $n!$, see Picard Schemes of Curves, Lemma 0B9G.

Let α_n be the zero cycle on X^n given by the formula in the statement of the lemma. Let $\mathcal{L} = \mathcal{O}_X(x-x')$. Then $c_1(\mathcal{L}) \cap [X] = [x] - [x']$. Thus

$$\alpha_n = c_1(\mathcal{L}_1) \cap \ldots \cap c_1(\mathcal{L}_n) \cap [X^n]$$

where $\mathcal{L}_i = \mathrm{pr}_i^*\mathcal{L}$ and $\mathrm{pr}_i : X^n \to X$ is the ith projection. By either Divisors, Lemma 0BD2 or Divisors, Lemma 0BD3 there is a norm for π. Set $\mathcal{N} = \mathrm{Norm}_\pi(\mathcal{L}_1)$, see Divisors, Lemma 0BCY. We have

$$\pi^*\mathcal{N} = (\mathcal{L}_1 \otimes \ldots \otimes \mathcal{L}_n)^{\otimes (n-1)!}$$

in $\mathrm{Pic}(X^n)$ by a calculation. Deails omitted; hint: this follows from the fact that $\mathrm{Norm}_\pi : \pi_*\mathcal{O}_{X^n} \to \mathcal{O}_{S^n(X)}$ composed with the natural map $\pi_*\mathcal{O}_{S^n(X)} \to \mathcal{O}_{X^n}$ is

equal to the product over all $\sigma \in S_n$ of the action of σ on $\pi_*\mathcal{O}_{X^n}$. Consider

$$\beta_n = c_1(\mathcal{N})^n \cap [S^n(X)]$$

in $\mathrm{CH}_0(S^n(X))$. Observe that $c_1(\mathcal{L}_i) \cap c_1(\mathcal{L}_i) = 0$ because $\mathcal{L}_i$ is pulled back from a curve, see Chow Homology, Lemma 0FDX. Thus we see that

$$\begin{aligned}
\pi^*\beta_n &= ((n-1)!)^n(\sum\nolimits_{i=1,\ldots,n} c_1(\mathcal{L}_i))^n \cap [X^n] \\
&= ((n-1)!)^n n^n c_1(\mathcal{L}_1) \cap \ldots \cap c_1(\mathcal{L}_n) \cap [X^n] \\
&= (n!)^n \alpha_n
\end{aligned}$$

Thus it suffices to show that β_n is torsion.

There is a canonical morphism

$$f : S^n(X) \longrightarrow \underline{\mathrm{Pic}}^n_{X/k}$$

See Picard Schemes of Curves, Lemma 0BA0. For $n \geq 2g-1$ this morphism is a projective space bundle (details omitted; compare with the proof of Picard Schemes of Curves, Lemma 0BA0). The invertible sheaf $\mathcal{N}$ is trivial on the fibres of f, see below. Thus by the projective space bundle formula (Chow Homology, Lemma 02TX) we see that $\mathcal{N} = f^*\mathcal{M}$ for some invertible module $\mathcal{M}$ on $\underline{\mathrm{Pic}}^n_{X/k}$. Of course, then we see that

$$c_1(\mathcal{N})^n = f^*(c_1(\mathcal{M})^n)$$

is zero because $n > g = \dim(\underline{\mathrm{Pic}}^n_{X/k})$ and we can use Chow Homology, Lemma 0FDX as before.

We still have to show that $\mathcal{N}$ is trivial on a fibre F of f. Since the fibres of f are projective spaces and since $\mathrm{Pic}(\mathbf{P}^m_k) = \mathbf{Z}$ (Divisors, Lemma 0BXJ), this can be shown by computing the degree of $\mathcal{N}$ on a line contained in the fibre. Instead we will prove it by proving that $\mathcal{N}$ is algebraically equivalent to zero. First we claim there is a connected finite type scheme T over k, an invertible module $\mathcal{L}'$ on $T \times X$ and k-rational points $p, q \in T$ such that $\mathcal{M}_p \cong \mathcal{O}_X$ and $\mathcal{M}_q = \mathcal{L}$. Namely, since $\mathcal{L} = \mathcal{O}_X(x - x')$ we can take $T = X$, $p = x'$, $q = x$, and $\mathcal{L}' = \mathcal{O}_{X\times X}(\Delta) \otimes \mathrm{pr}_2^*\mathcal{O}_X(-x')$. Then we let $\mathcal{L}'_i$ on $T \times X^n$ for $i = 1, \ldots, n$ be the pullback of $\mathcal{L}'$ by $\mathrm{id}_T \times \mathrm{pr}_i : T \times X^n \to T \times X$. Finally, we let $\mathcal{N}' = \mathrm{Norm}_{\mathrm{id}_T\times\pi}(\mathcal{L}'_1)$ on $T \times S^n(X)$. By construction we have $\mathcal{N}'_p = \mathcal{O}_{S^n(X)}$ and $\mathcal{N}'_q = \mathcal{N}$. We conclude that

$$\mathcal{N}'|_{T\times F}$$

is an invertible module on $T \times F \cong T \times \mathbf{P}^m_k$ whose fibre over p is the trivial invertible module and whose fibre over q is $\mathcal{N}|_F$. Since the euler characteristic of the trivial bundle is 1 and since this euler characteristic is locally constant in families (Derived Categories of Schemes, Lemma 0B9T) we conclude $\chi(F, \mathcal{N}^{\otimes s}|_F) = 1$ for all $s \in \mathbf{Z}$. This can happen only if $\mathcal{N}|_F \cong \mathcal{O}_F$ (see Cohomology of Schemes, Lemma 01XT) and the proof is complete. Some details omitted. □

References

[1] V. Voevodsky. A nilpotence theorem for cycles algebraically equivalent to zero. *Internat. Math. Res. Notices*, (4):187–198, 1995.